

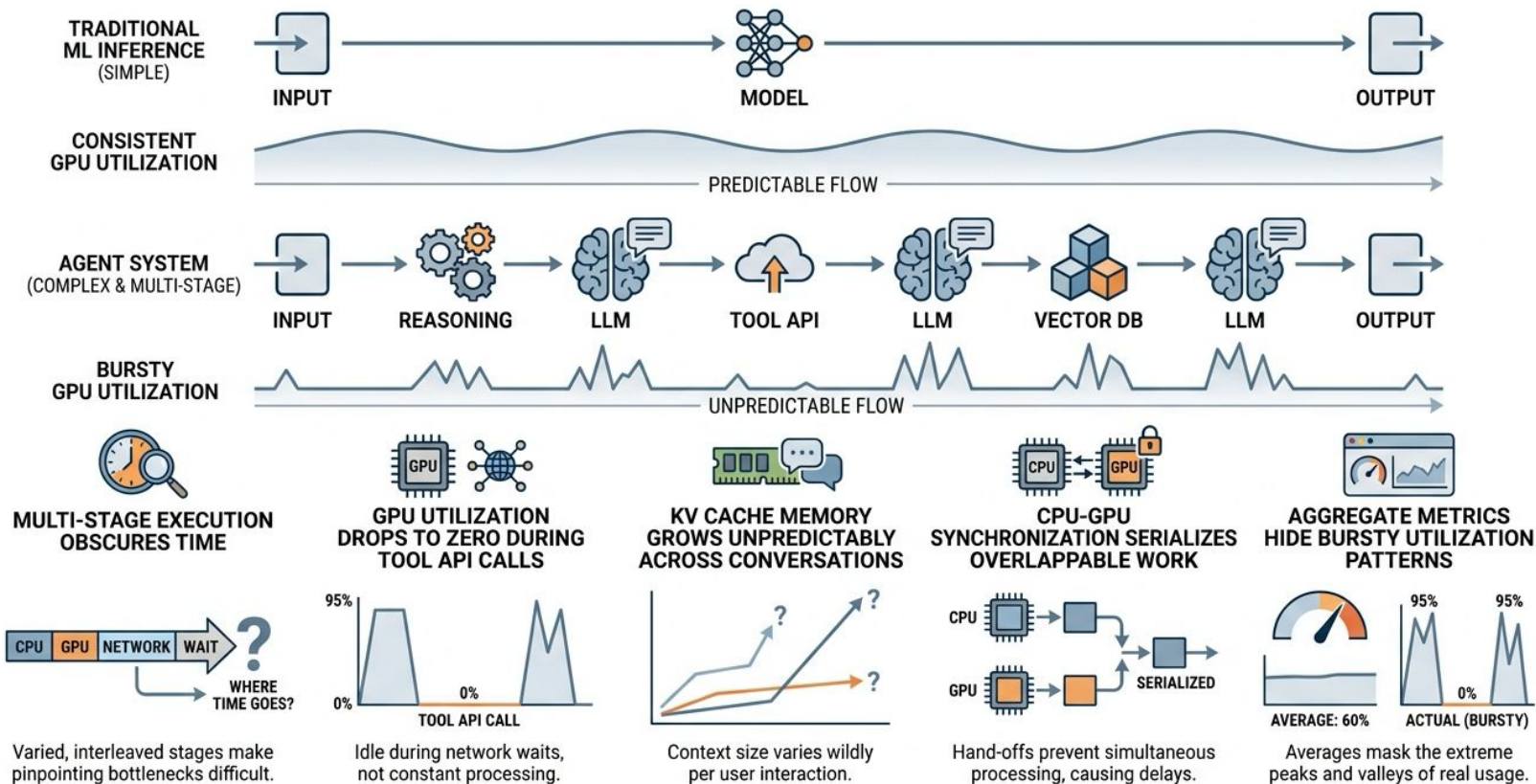


U.S. General Services
Administration

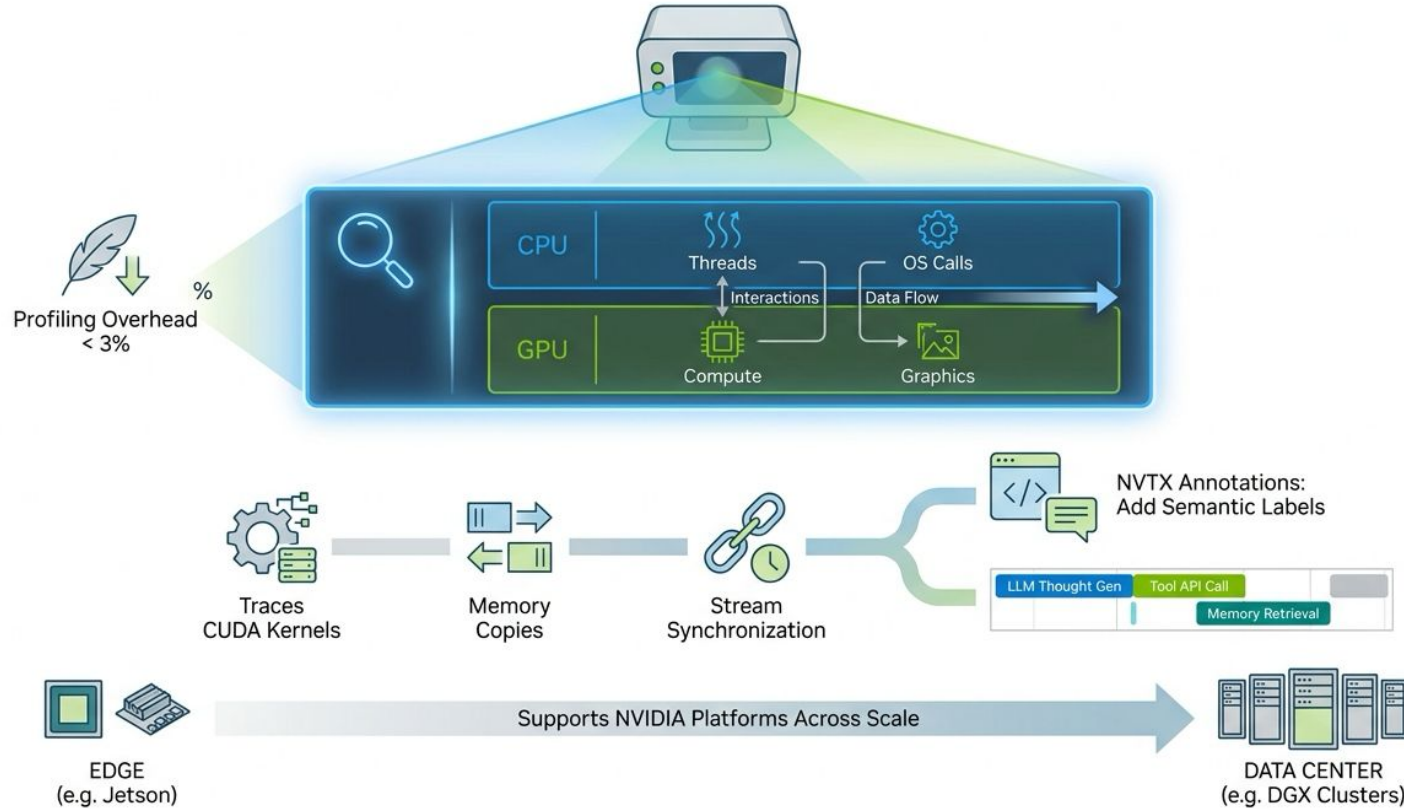
Chapter 4.4: Performance Profiling and Optimization

US General Service Administration
AI Community of Practice - March 2026

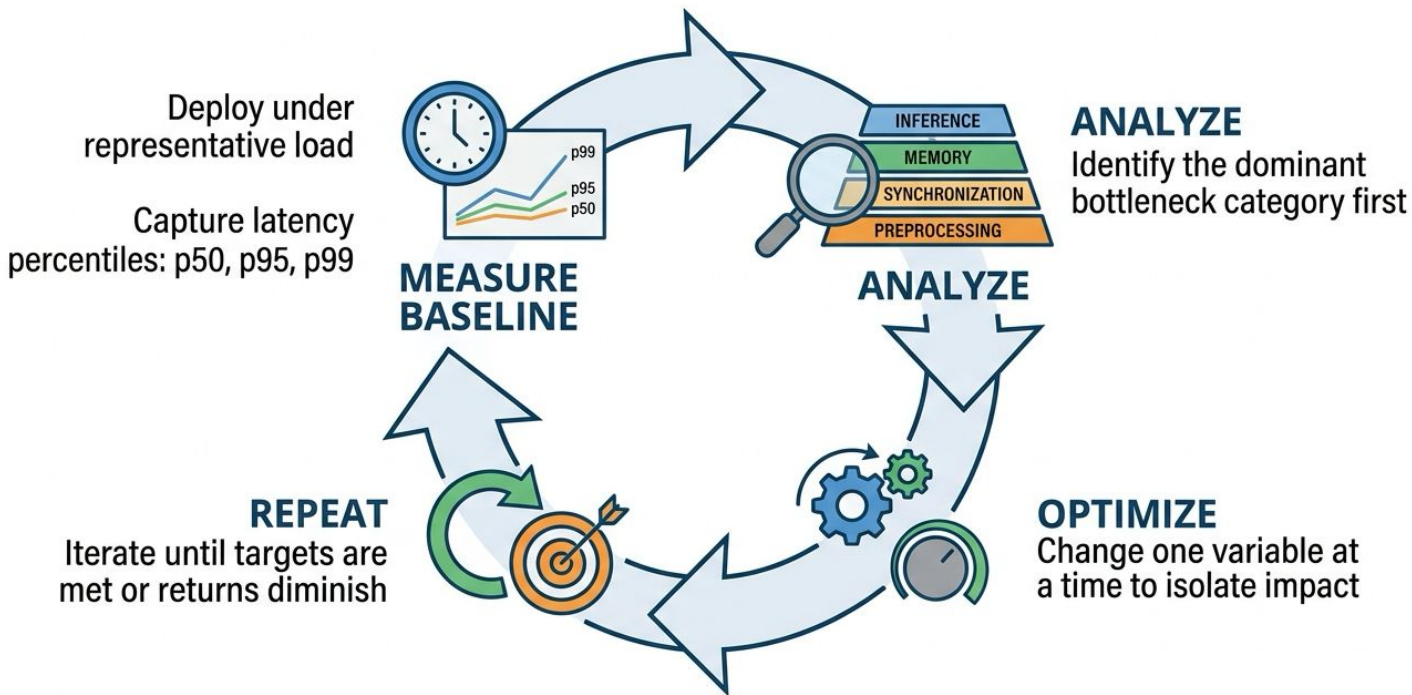
Why Agent Profiling Is Different



Nsight Systems -- Your Performance X-Ray

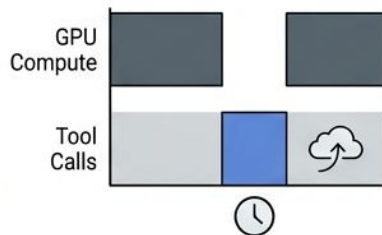


The Profiling Workflow

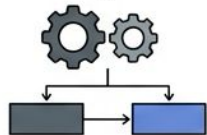


Reading Timeline Patterns

EXECUTION OVERLAP OPPORTUNITY



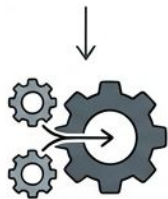
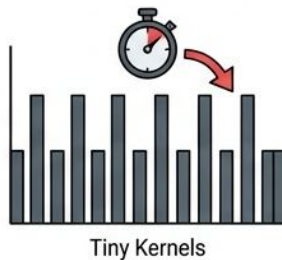
FIX



OVERLAP EXECUTION

GPU idle during tool calls signals need for concurrent processing

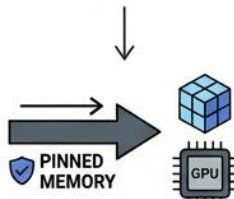
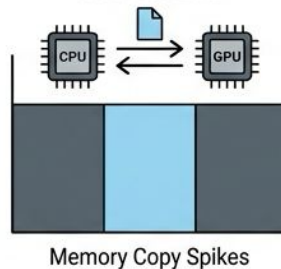
KERNEL OVERHEAD DOMINANCE



OPERATION FUSION / TORCHSCRIPT

Frequent small kernels indicate GIL or sync overhead; combine operations

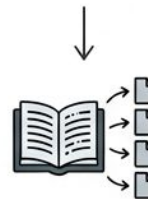
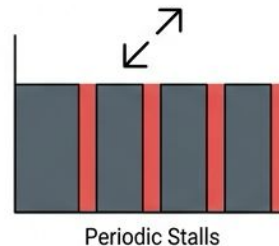
INEFFICIENT HOST-DEVICE TRANSFERS



OPTIMIZE DATA MOVEMENT

Large memory copy blocks reveal inefficient CPU-GPU transfers; use pinned memory, keep tensors on device

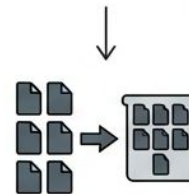
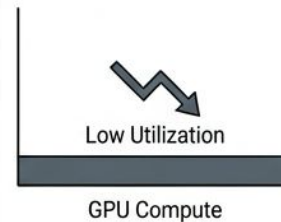
KV CACHE REALLOCATION PRESSURE



PAGED ATTENTION

Regular stalls suggest KV cache growth; allocate cache in small pages on demand

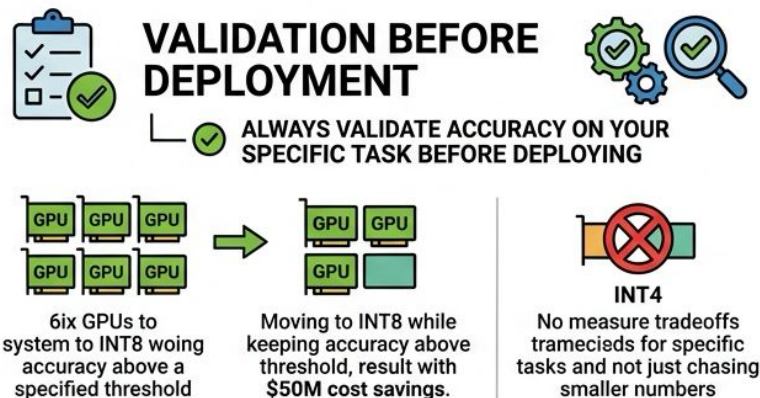
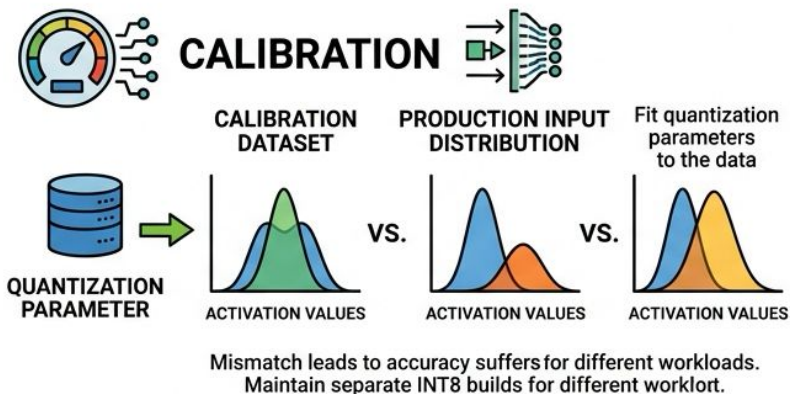
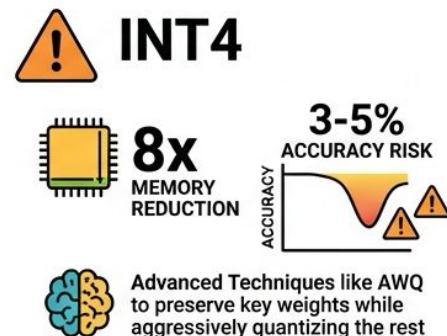
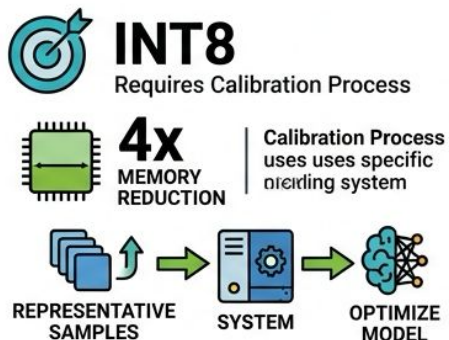
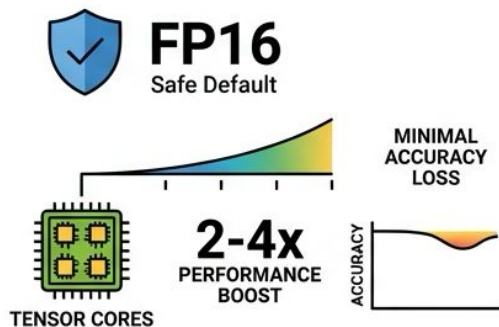
INSUFFICIENT BATCHING



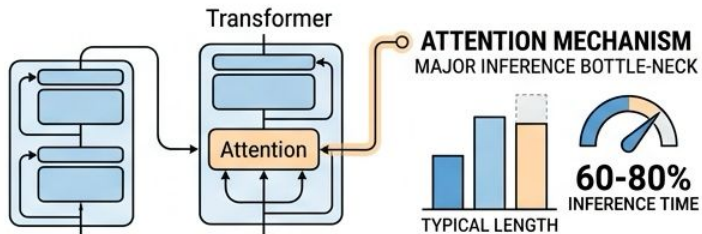
INCREASE BATCH SIZE

Low sustained utilization points to need for larger batch processing

Quantization -- Trading Precision for Performance

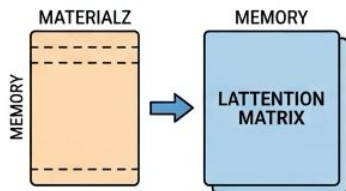


Attention Kernel Optimization

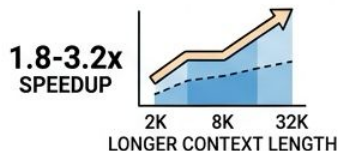


FLASH ATTENTION

STANDARD ATTENTION

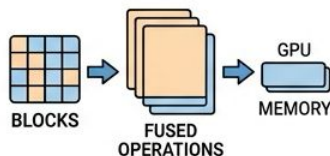


FULL ATTENTION SCORE MATRIX
TAKES A LOT OF GPU MEMORY



FLASH ATTENTION

ELIMINATES LARGE MATRIX STORAGE



SCORES BEING APPLIED IMMEDIATELY
MOVE THROUGH THE GPU MATRIX

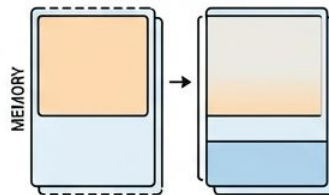


PAGED ATTENTION

FIXES KV CACHE FRAGMENTATION

STANDARD KV CACHE

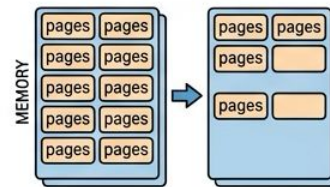
WASTED SPACES FOR
VARIABLE LENGTHS



WASTED RESERVATIONS
FOR VARIABLE LENGTHS

PAGED ATTENTION

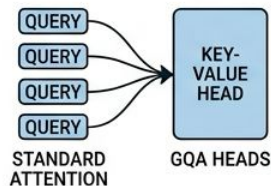
ALLOCATES SMALL, FIXED-SIZE
PAGES ON DEMAND



EFFICIENT UTILIZATION
FOR VARIABLE LENGTHS

GROUPED-QUERY ATTENTION (GQA)

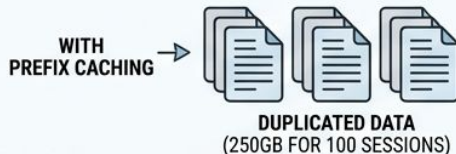
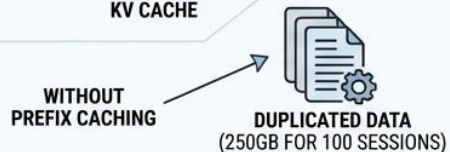
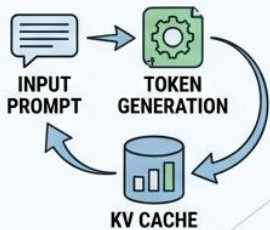
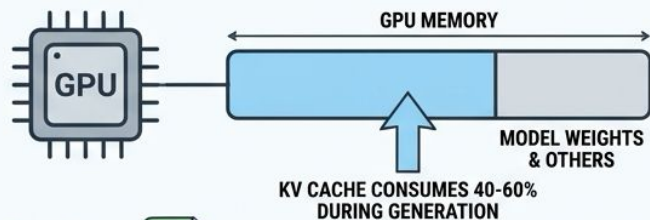
GQA IN
Llama-3



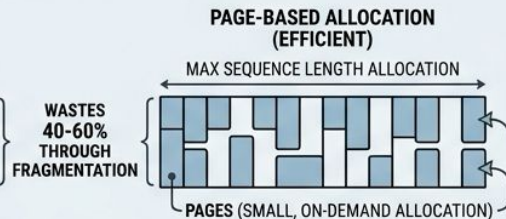
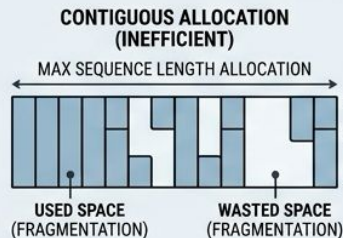
8x
KV CACHE
REDUCTION
NATIVELY

KV Cache Management and Prefix Caching

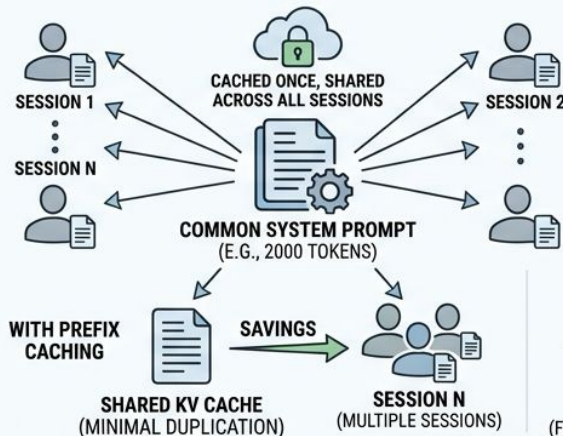
KV CACHE & GPU MEMORY CONSUMPTION



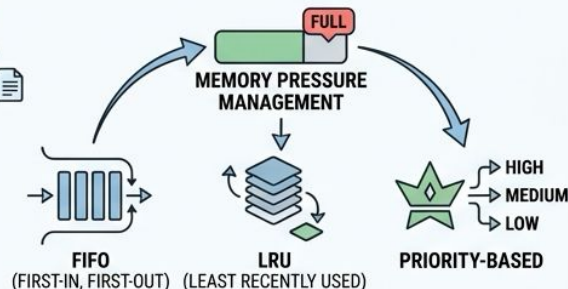
ALLOCATION & FRAGMENTATION



PREFIX CACHING & PROMPT SHARING

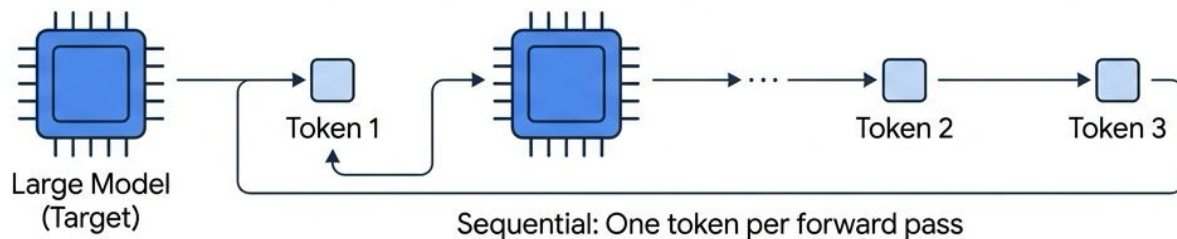


EVICION POLICIES (MEMORY PRESSURE)



Speculative Decoding -- Predicting the Predictable

Standard Generation

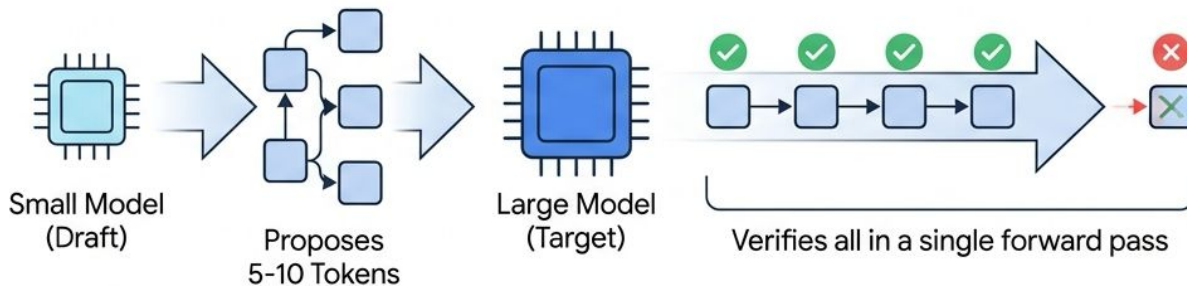


Throughput Improvement

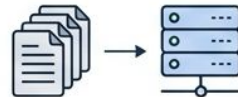


2.5-3.5x Throughput
(Moderately predictable outputs)

Speculative Decoding

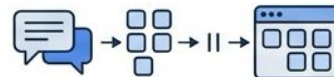


Best Use Case



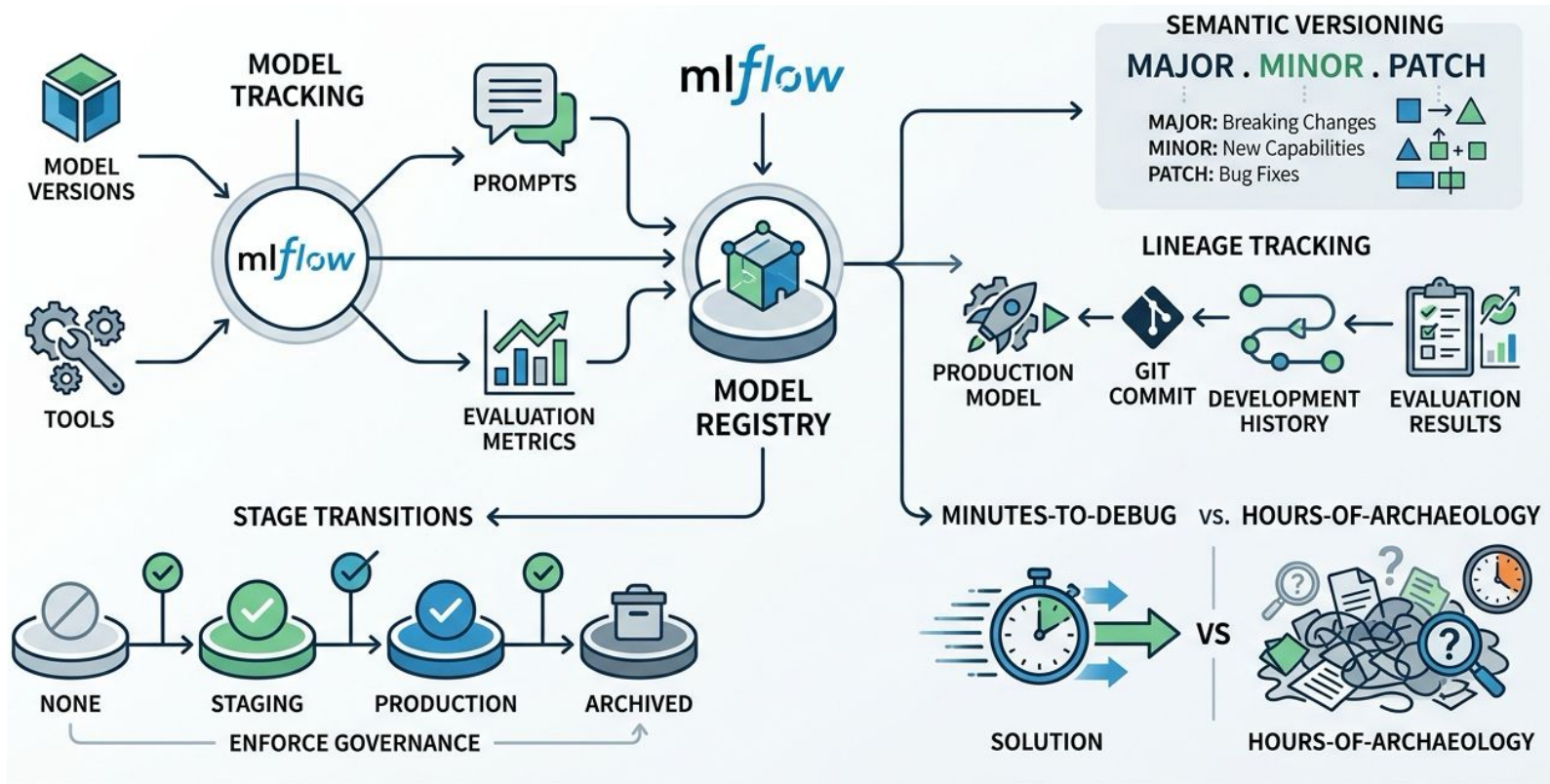
Batch Workloads

Less Ideal Use Case



Streaming Chat
(Irregular token bursts)

Model Registry -- Knowing What You Deployed



GitOps and Safe Rollbacks

- GitOps makes git the single source of truth for infrastructure
 - ArgoCD continuously reconciles cluster state to match git
 - Rollback is a git revert, not an emergency procedure
 - Canary deployments expose only 5% of users to new versions first
 - Automated rollback triggers on metric degradation within seconds
- 

Key Takeaways

- Profile first, optimize second -- data beats intuition
 - Nsight Systems plus NVTX annotations reveal the full picture
 - Quantization, Flash Attention, and speculative decoding stack together
 - MLflow plus GitOps provide artifact governance and safe deployments
 - Next chapter: NVIDIA NIM for production inference at scale
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